Exploring Variation in Gestation Length Across Eutherian Mammals

BIS015L_Winter 2025 Group10

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01 Background Information

02 Data Exploration

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01

Background Information

Goal:

Differential associations of gestation length with body mass and lifespan across mammals

Introduction of Dataset

- 845 representative extant species of eutherian mammals (covering approximately 14% of all known extant species)

Three Variables:

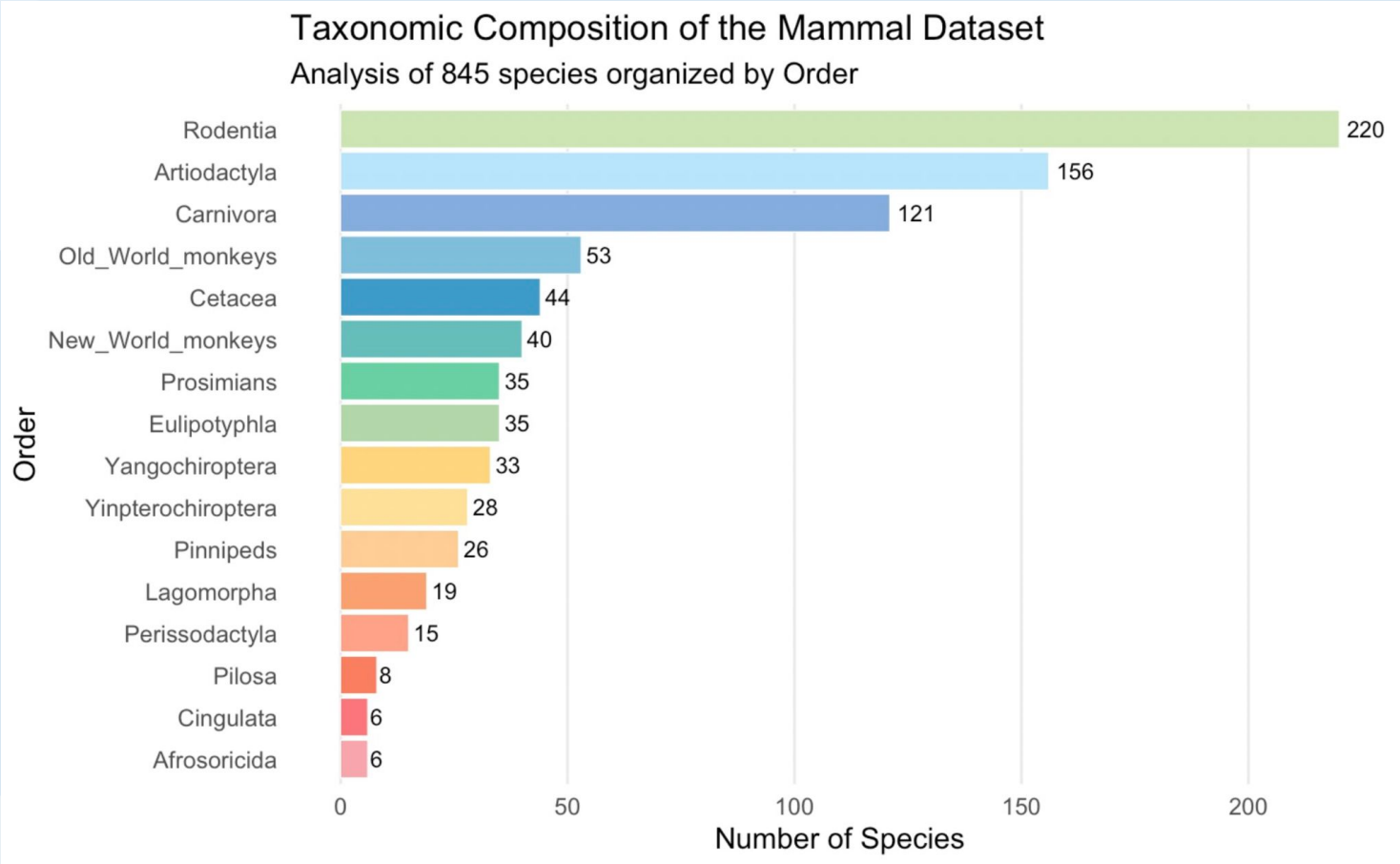
Body Mass: The body mass of each mammalian species, measured in grams (g)

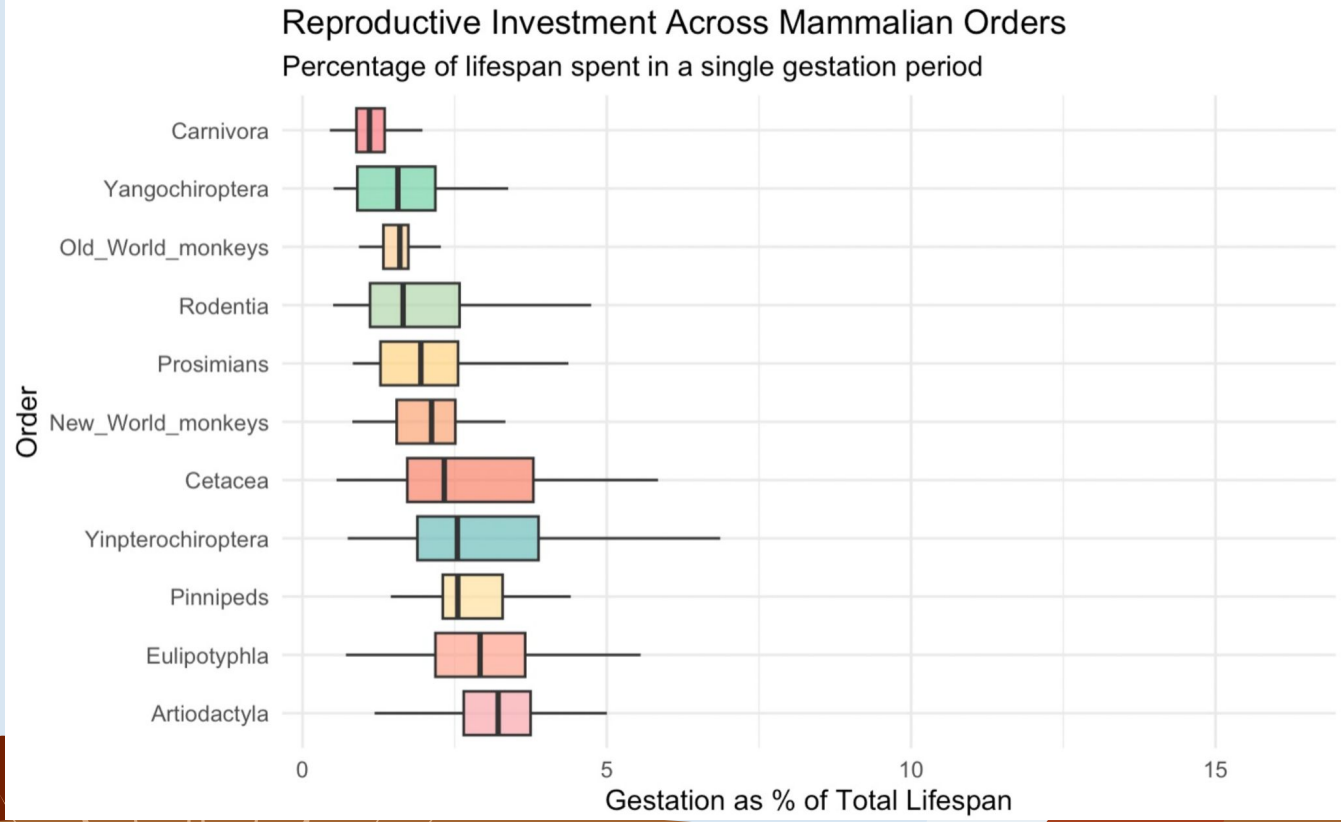
Longevity: The expected lifespan of each species, measured in years.

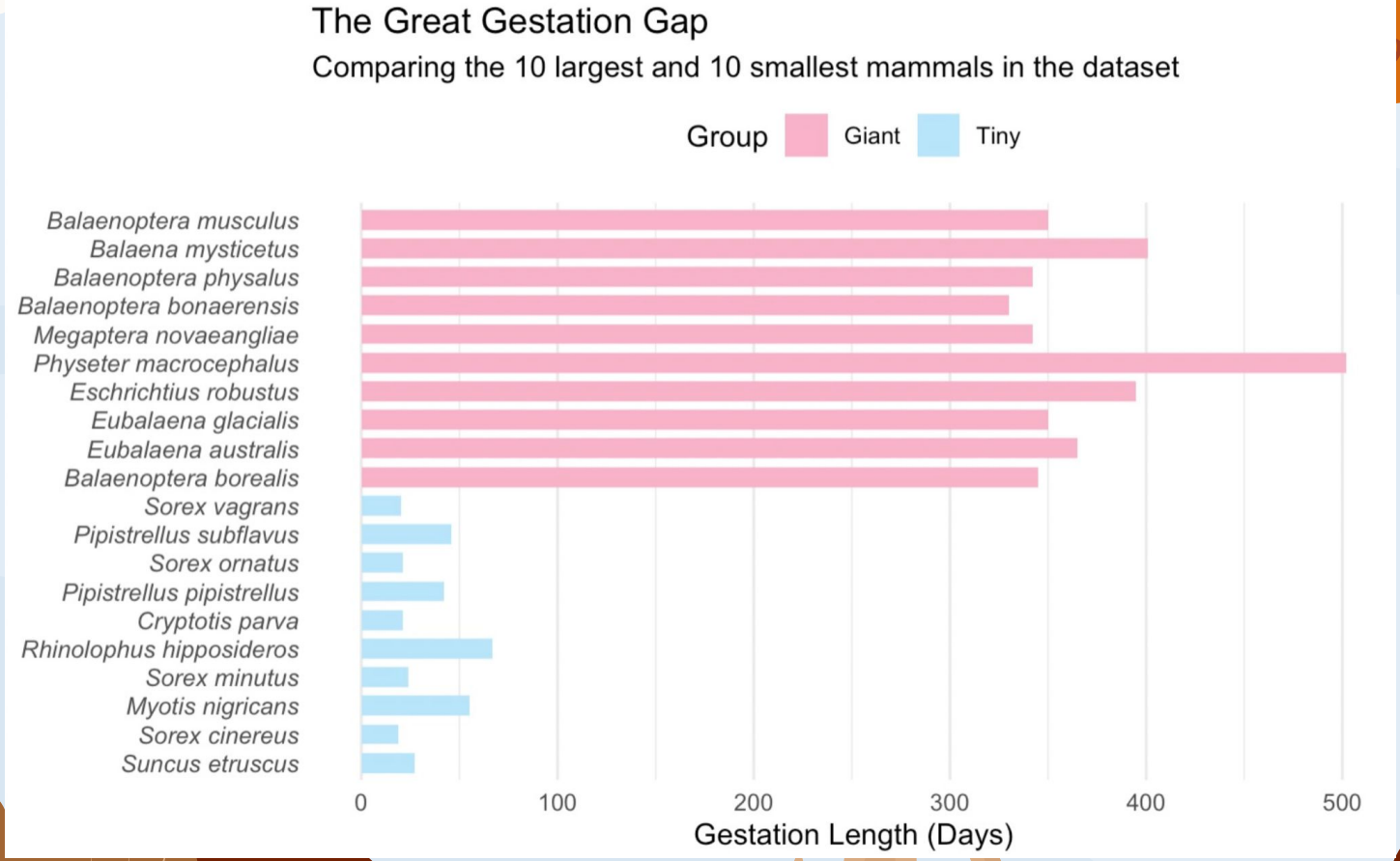
Gestation Length: The length of pregnancy for each species, measured in days

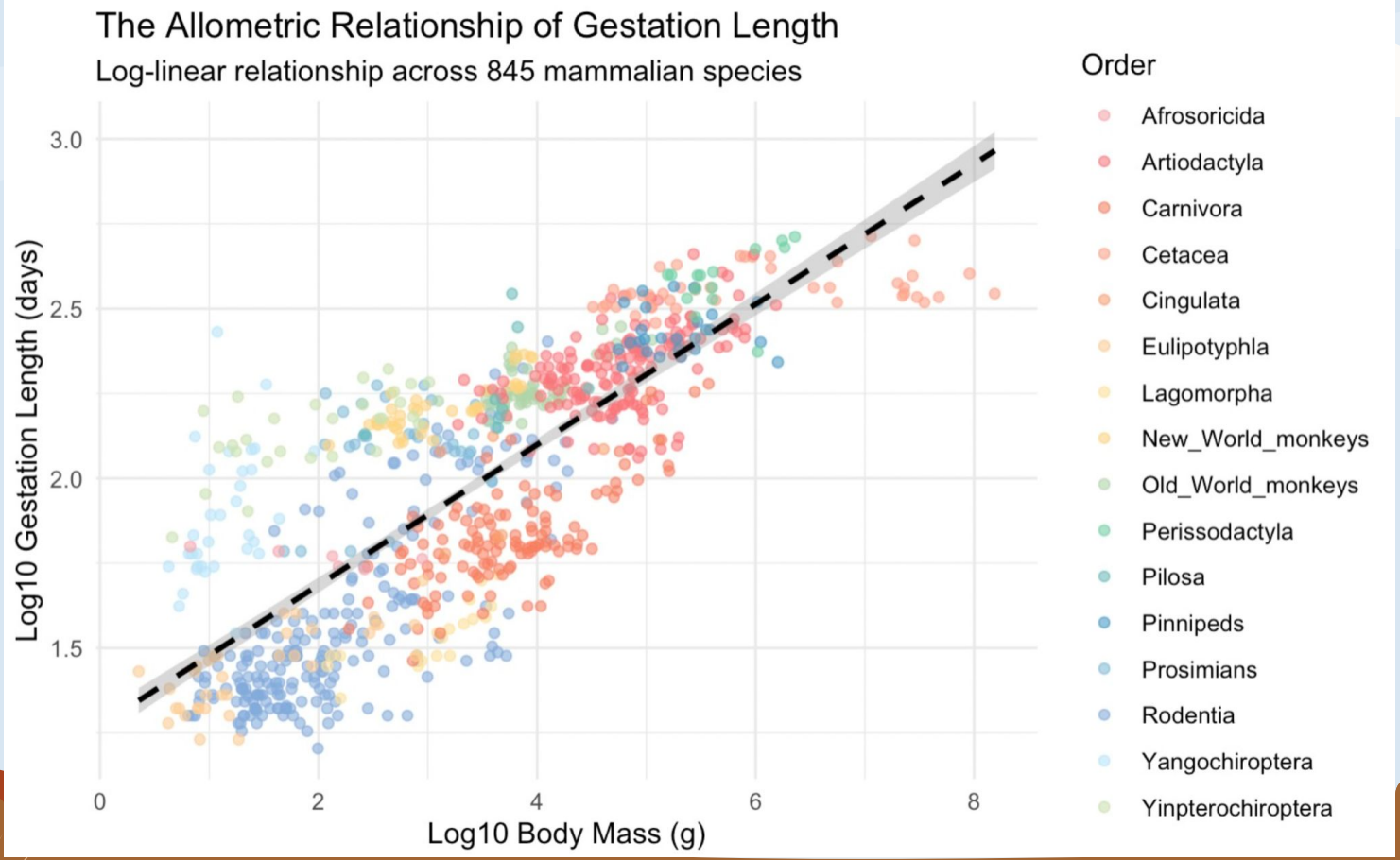
02

Data Exploration

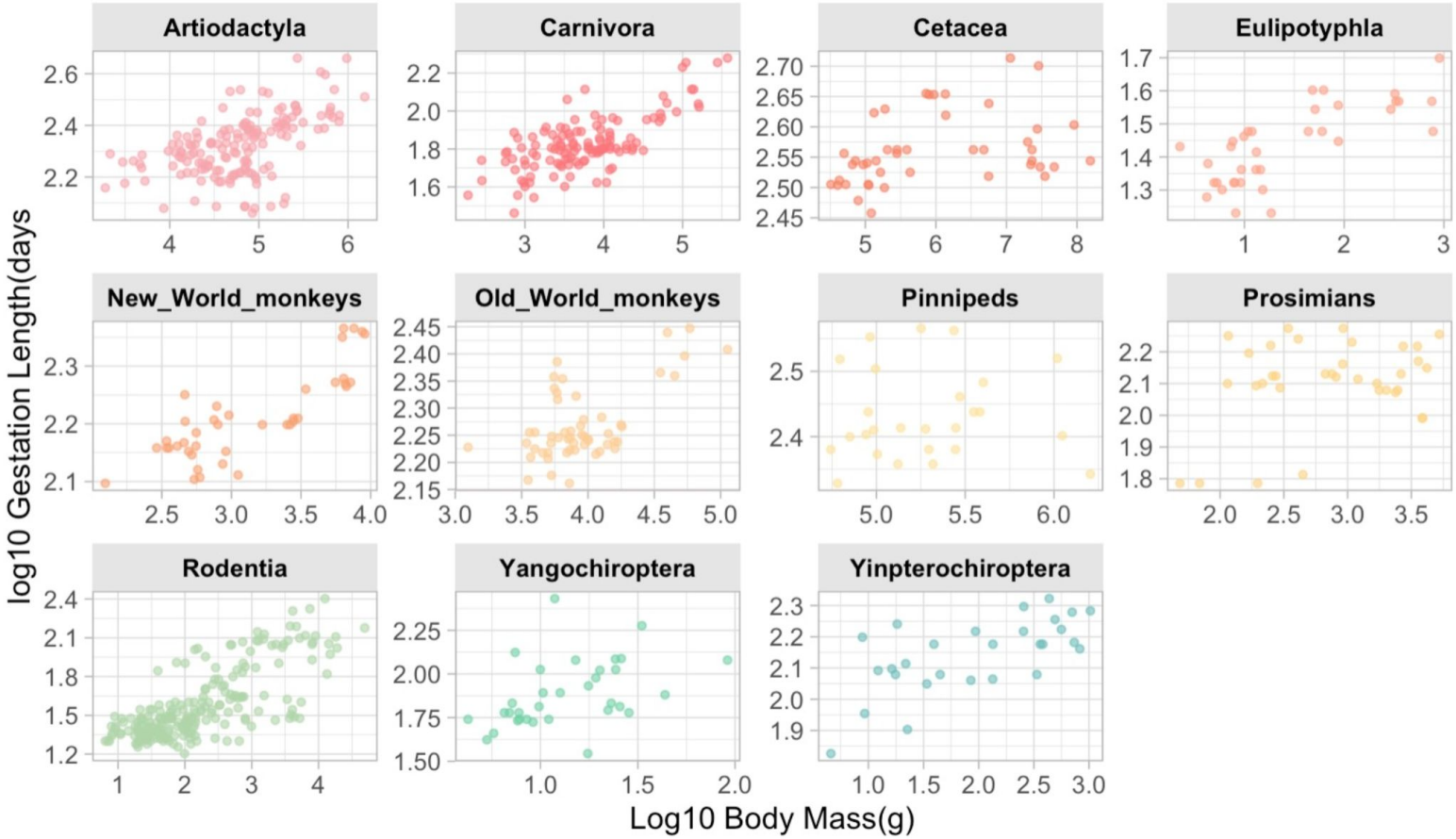

Taxonomic Composition of the Mammal Dataset
Analysis of 845 species organized by Order
Order
Rodentia
Artiodactyla
Carnivora
Old_World_monkeys
Cetacea
New_World_monkeys
Prosimians
Eulipotyphla
Yangochiroptera
Yinpterochiroptera
Pinnipeds
Lagomorpha
Perissodactyla
Pilosa
Cingulata
Afrosoricida
220
156
121
53
44
40
35
35
33
28
26
19
15
8
6
6
0
50
100
150
200
Number of Species

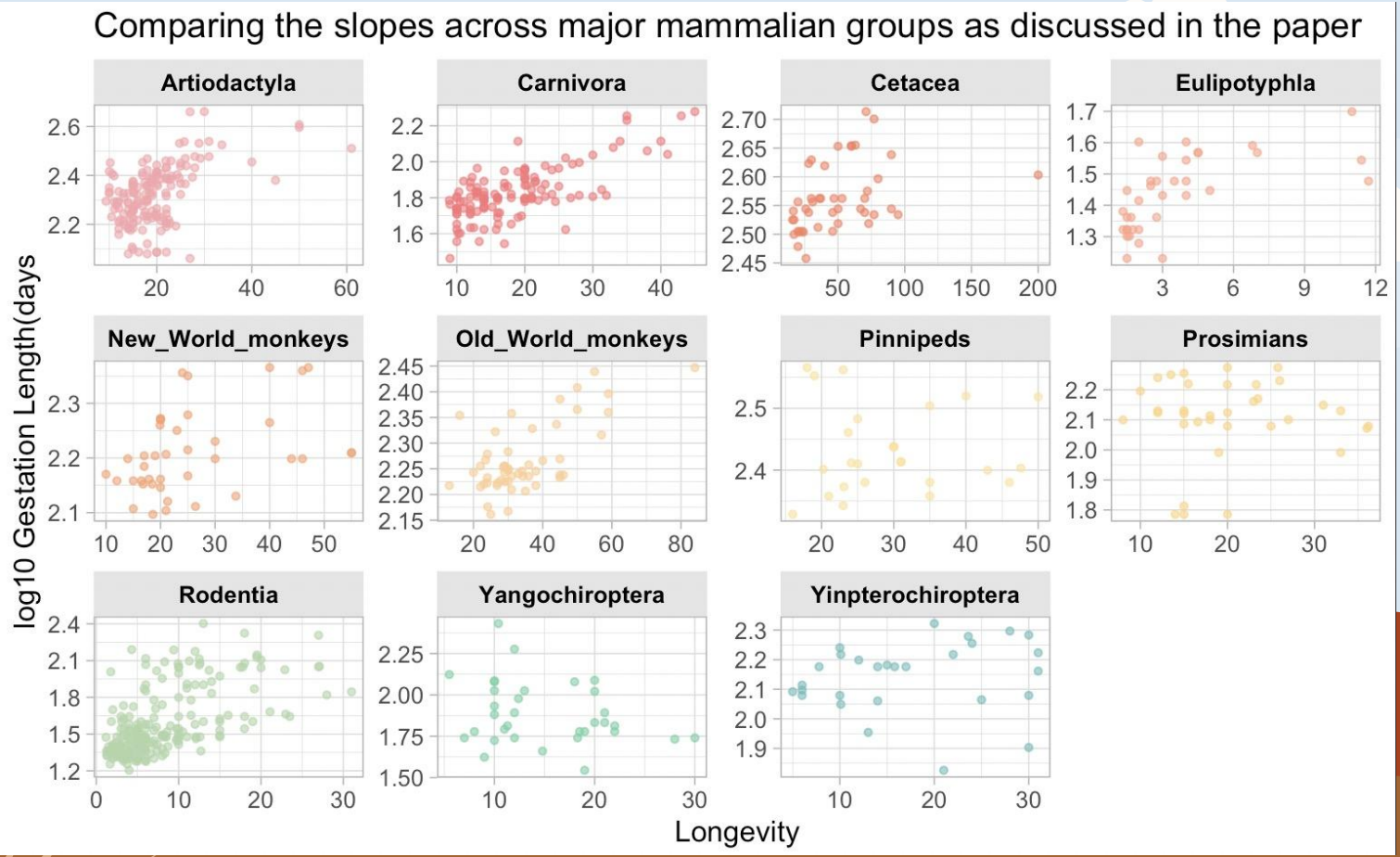

Reproductive Investment Across Mammalian Orders
Percentage of lifespan spent in a single gestation period
Order
Carnivora
Yangochiroptera
Old_World_monkeys
Rodentia
Prosimians
New_World_monkeys
Cetacea
Yinpterochiroptera
Pinnipeds
Eulipotyphla
Artiodactyla
0
5
10
15
Gestation as % of Total Lifespan


The Great Gestation Gap
Comparing the 10 largest and 10 smallest mammals in the dataset
Group
Giant
Tiny
Balaenoptera musculus
Balaena mysticetus
Balaenoptera physalus
Balaenoptera bonaerensis
Megaptera novaeangliae
Physeter macrocephalus
Eschrichtius robustus
Eubalaena glacialis
Eubalaena australis
Balaenoptera borealis
Sorex vagrans
Pipistrellus subflavus
Sorex ornatus
Pipistrellus pipistrellus
Cryptotis parva
Rhinolophus hipposideros
Sorex minutus
Myotis nigricans
Sorex cinereus
Suncus etruscus
0
100
200
300
400
500
Gestation Length (Days)


The Allometric Relationship of Gestation Length
Log-linear relationship across 845 mammalian species
Log10 Gestation Length (days)
3.0
2.5
2.0
1.5
0
2
4
6
8
Log10 Body Mass (g)
Order
Afrosoricida
Artiodactyla
Carnivora
Cetacea
Cingulata
Eulipotyphla
Lagomorpha
New_World_monkeys
Old_World_monkeys
Perissodactyla
Pilosa
Pinnipeds
Prosimians
Rodentia
Yangochiroptera
Yinpterochiroptera

Comparing the slopes across major mammalian groups as discussed in the paper




Comparing the slopes across major mammalian groups as discussed in the paper
Artiodactyla
Carnivora
Cetacea
Eulipotyphla
New_World_monkeys
Old_World_monkeys
Pinnipeds
Prosimians
Rodentia
Yangochiroptera
Yinpterochiroptera
log10 Gestation Length(days
Longevity

Conclusion:

Gestation length varies widely among mammals and has evolved independently in different lineages.
Its relationship with body mass and lifespan differs across orders

03

Shinyapp

Thanks!